

Module 8 Lab:
pfSense Firewall Analysis
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Introduction

Firewalls play a central role in modern network security by controlling how traffic flows between trusted and untrusted environments. In this lab, pfSense functions as the firewall and router between multiple internal networks—LAN1, LAN2, and the WAN interface—allowing us to observe how rule ordering, interface selection, and traffic direction influence security outcomes. The Metasploitable virtual machine, intentionally designed with numerous vulnerabilities, provides a realistic target for understanding how firewalls protect high-risk systems. This paper answers two key questions: how to protect a vulnerable host like Metasploitable using pfSense, and how to design firewall rules that secure an enterprise network (LAN1) from an attacker network (LAN2) while still allowing legitimate traffic to pass.

49. What firewall rules can we set up in pfSense to protect the Metasploitable VM?

Metasploitable is intentionally insecure, containing outdated services, weak configurations, and known vulnerabilities. Because of this, it should be treated as a high-risk asset that requires strict isolation. pfSense provides several mechanisms to enforce this isolation through interface-specific rules and default-deny behavior.

The first and most important rule is to **block all inbound traffic from the WAN to the Metasploitable IP address**. Although pfSense blocks unsolicited inbound WAN traffic by default, creating an explicit block rule reinforces the security posture and prevents accidental exposure. This rule would specify the WAN interface as the entry point, “any” as the source, and the Metasploitable IP (for example, 10.0.1.3) as the destination. This ensures that no external system can initiate communication with the vulnerable host.

The second layer of protection involves **restricting internal access**. Even within a private network, not every device should be able to reach Metasploitable. pfSense allows

administrators to create granular rules that specify which internal hosts or subnets are permitted to communicate with it. For example, a rule on the LAN1 interface could allow only a designated testing machine—such as Kali-LAN1—to access Metasploitable. This rule would be followed by a broader deny rule that blocks all other LAN1 devices from reaching the Metasploitable IP. Because pfSense processes rules from top to bottom, the specific allow rule must appear above the general deny rule.

Together, these rules create a controlled environment where Metasploitable is reachable only by authorized internal systems and completely isolated from the Internet. This approach mirrors real-world practices for isolating legacy or high-risk systems that cannot be fully secured but still need to be accessible for testing or research.

50. How do you protect LAN1 (enterprise network) from LAN2 (attacker network) while still allowing good requests through?

In this scenario, LAN1 represents an enterprise network containing valuable resources, while LAN2 represents an untrusted or attacker-controlled network. The goal is to prevent unwanted or malicious traffic from LAN2 from reaching LAN1, while still allowing legitimate communication when necessary. This reflects common enterprise segmentation strategies, where internal networks are divided into trust zones and traffic between them is tightly controlled.

The most important principle in this situation is **default deny**. pfSense allows administrators to enforce this by creating a rule on the OPT1 (LAN2) interface that blocks all traffic from LAN2 to LAN1. This rule establishes a protective boundary: no device on the attacker network can reach any system on the enterprise network unless explicitly permitted. This mirrors how organizations isolate guest Wi-Fi networks, contractor networks, or lab environments from production systems.

Once the default deny posture is in place, administrators can selectively allow “good” traffic. For example, if a web server in LAN1 needs to be accessible from LAN2 for testing purposes, a narrow allow rule can be added above the block rule. This rule would specify LAN2 as the source, the web server’s IP address as the destination, and TCP port 443 as the allowed service. By limiting the rule to a single host and a single protocol, the administrator minimizes the attack surface while still enabling legitimate use cases.

This layered approach—deny broadly, allow narrowly—ensures that enterprise resources remain protected while still supporting necessary workflows. pfSense’s rule ordering, interface-specific policies, and ability to match on protocol, port, and subnet make it well-suited for implementing this type of segmentation.

Conclusion

pfSense provides a powerful and flexible framework for protecting both vulnerable systems and enterprise networks. In the case of Metasploitable, pfSense enables administrators to isolate the host from the Internet and restrict internal access to only authorized testing machines. This aligns directly with the course’s framing of firewalls as “customs agents” at the network border—inspecting every packet, determining its purpose, and blocking anything that does not meet clearly defined criteria. By applying interface-specific rules and carefully ordering them, pfSense enforces these decisions with precision.

More broadly, the lab demonstrates the importance of layered defense. Firewalls, segmentation, and access controls work together to create multiple checkpoints between untrusted and trusted environments. This mirrors the perimeter-security model emphasized in the course, where no single control is sufficient on its own. Instead, each layer—WAN filtering,

LAN segmentation, rule ordering, and explicit allow-lists—contributes to a cohesive security posture.

Across both use cases in this lab, the underlying principle remains consistent: start with deny everything from untrusted to trusted, then add narrow, explicit allow rules for only the traffic you truly intend to permit (by source, destination, protocol, and port). This approach reflects real-world enterprise security practices and forms the foundation of effective network protection against both external threats and internal misuse.

References

Sawant, A. (2026). Perimeter Network Security. [Online course]. Coursera & Northeastern University.

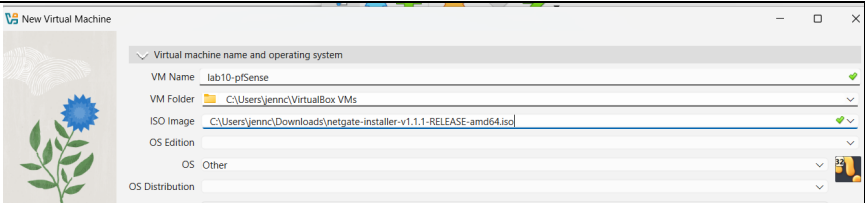
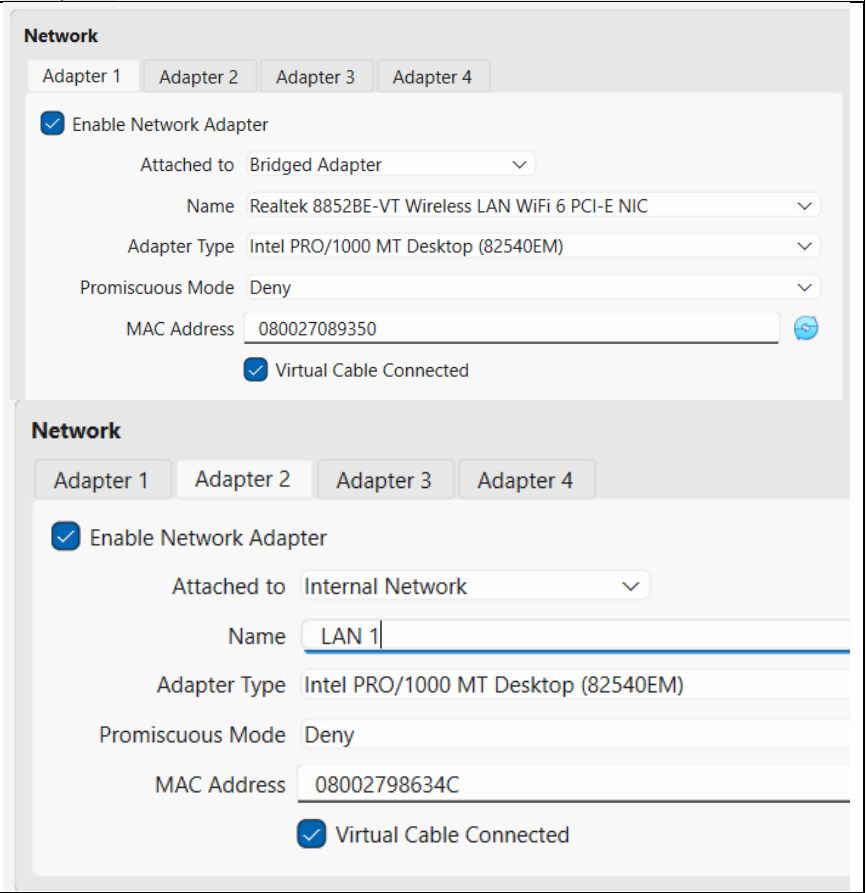
Methodology and Use of AI Tools

I completed all technical steps of this lab directly in the Kali Linux virtual machine, including packet capture, filter creation, profile configuration, and detailed analysis of TLS, DNS, and HTTP/OCSP traffic. All screenshots and observations were generated from my own live capture sessions. I used Microsoft Copilot as a writing and organizational aid to help structure the report, refine transitions between sections, and ensure alignment with the assignment rubric. Copilot did not perform any technical steps in Wireshark or interpret packet data on my behalf. All analysis, interpretations, and conclusions reflect my own understanding of the material and the concepts covered in Module 7.

APPENDIX: Documented Process

The following table documents the exact steps performed for this lab, with each left-column description corresponding to the screenshot or output shown in the right column.

Before capturing screenshots, I also verified that the OPT1 block rule was functioning by checking pfSense’s firewall logs (Status → System Logs → Firewall). The logs confirmed that ICMP packets originating from LAN2 were being blocked after the rule was applied.

Step 1: Download & install pfSense iso on a new VM	
Step 2: Update Network settings and boot order	

Network

Adapter 1 Adapter 2 Adapter 3 Adapter 4

☒ Enable Network Adapter

Attached to Internal Network

Name LAN 2

Adapter Type Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode Deny

MAC Address 08002782D280

☒ Virtual Cable Connected

System

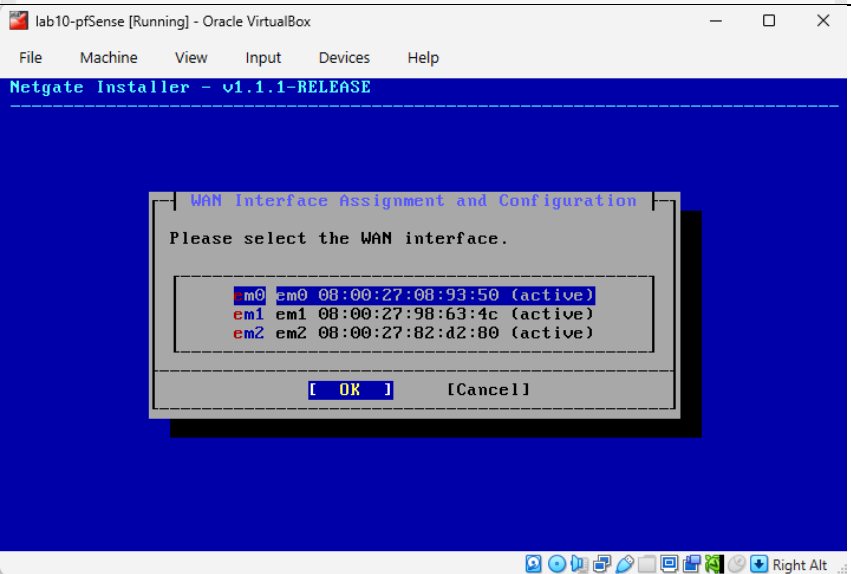
Motherboard Processor Acceleration

Base Memory 2048 MB

Boot Device Order (BIOS only)

- ☒ Hard Disk
- ☒ Optical
- ☒ Floppy
- ☐ Network

Step 3: Complete install
of pfSense



Selecting the WAN that matches Adapter 1

```
lab10-pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Installation Details

Installing pfSense base:

Updating pfSense-core repository catalogue...
pkg-static: Warning: Major OS version upgrade detected. Running "pkg boot
Fetching meta.conf:
Fetching data.pkg:
pfSense-core repository is up to date.
Updating pfSense repository catalogue...
Fetching meta.conf:
Fetching data.pkg:
pfSense repository is up to date.
All repositories are up to date.
The following 1 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
  pfSense-base: 2.8.1 [pfSense-core]

Number of packages to be installed: 1

The process will require 104 MiB more space.
103 MiB to be downloaded.
```

Downloading and installing the files

```
lab10-pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Netgate Installer - v1.1.1-RELEASE

Complete

Installation of pfSense complete! Would you
like to reboot into the installed system now?

[Reboot] [Shell]
```

PART 3: Configure pfSense

Step 4: assign LAN interface options

```
lab10-pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Generating RRD graphs...done.
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20251215-1731
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 4fc73e0578810f93e68e

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.14/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults         13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

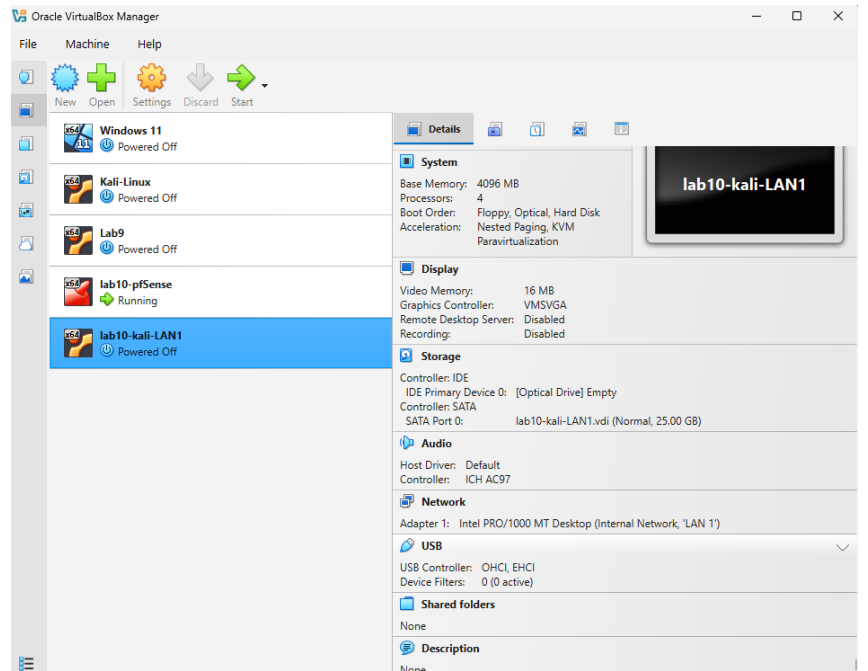
Enter an option: 
```

	lab10-pfSense [Running] - Oracle VirtualBox FileMachineViewInputDevicesHelp Valid interfaces are: em0 08:00:27:08:93:50 (up) Intel(R) Legacy PRO/1000 MT 82540EM em1 08:00:27:98:63:4c (down) Intel(R) Legacy PRO/1000 MT 82540EM em2 08:00:27:82:d2:80 (down) Intel(R) Legacy PRO/1000 MT 82540EM Do VLANs need to be set up first? If VLANs will not be used, or only for optional interfaces, it is typical to say no here and use the webConfigurator to configure VLANs later, if required. Should VLANs be set up now [y/n]? n If the names of the interfaces are not known, auto-detection can be used instead. To use auto-detection, please disconnect all interfaces before pressing 'a' to begin the process. Enter the WAN interface name or 'a' for auto-detection (em0 em1 em2 or a): em0 Enter the LAN interface name or 'a' for auto-detection NOTE: this enables full Firewalling/NAT mode. (em1 em2 a or nothing if finished): em1 Enter the Optional 1 interface name or 'a' for auto-detection (em2 a or nothing if finished): em2 The interfaces will be assigned as follows: WAN -> em0 LAN -> em1 OPT1 -> em2 Do you want to proceed [y/n]?
Step 5: Configure LAN 1 and LAN 2	IP Ranges updated: WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.14/24 LAN (lan) -> em1 -> v4: 10.0.1.1/24 OPT1 (opt1) -> em2 -> v4: 10.0.3.1/24

Part 4: Create VMs into LAN1 and LAN2

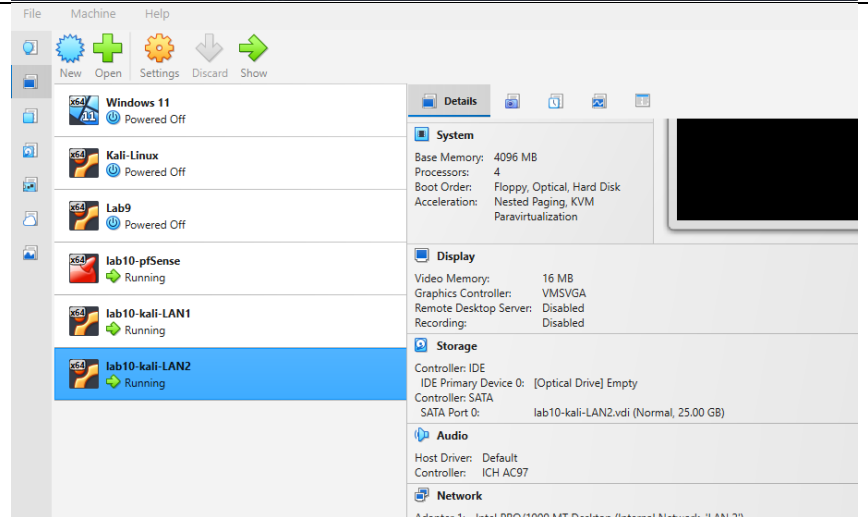
Step 6: VM clone for LAN1 and check to see if it is getting IP address from pfSense

VM for LAN1:



```
(jenn@kali-linux)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:ef:88:b0 brd ff:ff:ff:ff:ff:ff
    inet 10.0.1.2/24 brd 10.0.1.255 scope global dynamic noprefixroute eth0
        valid_lft 6821sec preferred_lft 6821sec
    inet6 fe80::a00:27ff:feef:88b0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Step 7: VM clone for LAN2 and check to see if it is getting IP address from pfSense



	<pre>(jenn@kali-linux)-[~] \$ ip a 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000 link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 inet 127.0.0.1/8 scope host lo valid_lft forever preferred_lft forever inet6 ::1/128 scope host noprefixroute valid_lft forever preferred_lft forever 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000 link/ether 08:00:27:6a:33:aa brd ff:ff:ff:ff:ff:ff inet 10.0.3.2/24 brd 10.0.3.255 scope global dynamic noprefixroute eth0 valid_lft 7171sec preferred_lft 7171sec inet6 fe80::a00:27ff:fe6a:33aa/64 scope link noprefixroute valid_lft forever preferred_lft forever</pre>
Step 8: Clone metasploitable and update network settings, start it up and confirm IP address	metasploitable-LAN1 - Settings BasicExpert GeneralSystemDisplayStorageAudioNetworkSerial PortsUSBShared FoldersUser Interface General IdentityFeaturesDescriptionDisk Encryption VM Name: metasploitable-LAN1 OS: Linux OS Distribution: Debian OS Version: Debian (64-bit) System MotherboardProcessorAcceleration Base Memory: 2048 MB 4 MB16384 MB Boot Device Order (BIOS only): FloppyOpticalHard Disk OKCancelHelp <pre>msfadmin@metasploitable:~\$ ip a 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 inet 127.0.0.1/8 scope host lo inet6 ::1/128 scope host valid_lft forever preferred_lft forever 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000 link/ether 08:00:27:b7:97:2f brd ff:ff:ff:ff:ff:ff inet 10.0.1.3/24 brd 10.0.1.255 scope global eth0 inet6 fe80::a00:27ff:feb7:972f/64 scope link valid_lft forever preferred_lft forever msfadmin@metasploitable:~\$</pre>
Part 5: Use Web GUI to configure pfSense Step 9: complete pfSense setup	Wizard / pfSense Setup / Wizard completed. Step 9 of 9 Wizard completed. Congratulations! pfSense is now configured. We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network. Check for updates Remember, we're here to help. Click here to learn about Netgate 24/7/365 support services. User survey Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous) Anonymous User Survey

Step 10: Take note of current IP addresses

Firewall / Rules / OPT1

The firewall rule configuration has been changed.
The changes must be applied for them to take effect.

✓ Apply Changes

Floating
WAN
LAN
OPT1

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 *	OPT1 address	*	LAN subnets	*	*	none	Allow traffic from LAN 2 to LAN 1	

↑ Add
↓ Add
Delete
Toggle
Copy
Save
+ Separator

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗	0/0 B	IPv4 ICMP	OPT1 address	*	LAN subnets	*	*	none		
			any								
☐	✓	0/0 B	IPv4 *	OPT1 address	*	LAN subnets	*	*	none	Allow traffic from LAN 2 to LAN 1	

↑ Add
↓ Add
Delete
Toggle
Copy
Save
+ Separator

Step 11: Configure a rule to allow and block traffic from LAN2 to LAN1

Step 12: Test again with Ping – this screen cap shows both tests with failures

```
jenn@kali-linux: ~  
Session Actions Edit View Help  
└─$ ping 10.0.1.3  
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.  
^C  
— 10.0.1.3 ping statistics —  
80 packets transmitted, 0 received, 100% packet loss, time 80972ms  
  
(jenn@kali-linux)-[~]  
└─$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:6a:33:aa brd ff:ff:ff:ff:ff:ff  
    inet 10.0.3.2/24 brd 10.0.3.255 scope global dynamic noprefixroute eth0  
        valid_lft 6804sec preferred_lft 6804sec  
    inet6 fe80::a00:27ff:fe6a:33aa/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(jenn@kali-linux)-[~]  
└─$ ping 10.0.1.3  
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.  
^C  
— 10.0.1.3 ping statistics —  
337 packets transmitted, 0 received, 100% packet loss, time 345310ms  
  
(jenn@kali-linux)-[~]  
└─$
```